National 5 Engineering Science

Yenka Practical Tasks – Analogue Electronics

🧪 Task 1 – Verifying Ohm's Law

Learning Intentions

- To measure voltage and current in a simple circuit
- To verify Ohm's Law experimentally

Instructions

1. Build a simple circuit:
    - DC power supply (set to 6V)
    - 100Ω resistor
    - Ammeter (series)
    - Voltmeter (across resistor)
2. Record:
    - Supply voltage
    - Current
    - Measured voltage across resistor
3. Calculate expected current using:

$$I = \frac{V}{R}$$

4. Compare calculated and simulated values.

Extension

Change the supply to:

- 3V
- 9V
- 12V

Record results in a table and comment on the pattern.

🔌 Task 2 – Series Circuits Investigation

Learning Intentions

- To investigate voltage sharing in series circuits

Build:

- 9V supply
- 220Ω resistor
- 470Ω resistor
- Ammeter in series
- Voltmeters across each resistor

Pupils must:

1. Measure total current.
2. Measure voltage across each resistor.
3. Confirm:
    - Voltage adds to supply voltage.
    - Current is the same everywhere.
4. Compare results with calculated values and discuss any issues found.

Challenge

Swap resistor values and predict results before simulating.

⚡ Task 3 – Parallel Circuits Investigation

Learning Intentions

- To explore current splitting in parallel circuits

Build:

- 12V supply
- 100Ω resistor (placed in parallel)
- 300Ω resistor
- Ammeter in main line
- Ammeters in each branch

Pupils must:

1. Measure branch currents.
2. Confirm:
    - Total current = sum of branch currents.

    - Voltage across each branch is equal.

Extension

Add a third resistor and predict the effect on:

- Total resistance
- Total current

🔀 Task 4 – Combined Circuit Challenge

Build:

- 100Ω resistor in series
- 400Ω and 600Ω resistors in parallel
- 20V supply

Tasks

1. Calculate total resistance before building.
2. Build circuit in Yenka.
3. Measure:
    - Total current
    - Voltage across parallel section
4. Compare calculated vs simulated results.

Reflection Question

Why is the current through the 400Ω resistor greater than through the 600Ω resistor?

🛠 Task 5 – Design Challenge

Scenario

A student needs 50Ω but only has two 100Ω resistors.

Task

1. Use Yenka to create a 50Ω equivalent resistance.
2. Prove using measurement tools.
3. Screenshot and annotate their circuit.
4. Explain why the solution works.

Extension

Create:

- 25Ω using three 100Ω resistors.

- 75Ω using three 100Ω resistors.

📊 Task 6 – Practical Data Task (Assessment Style)

Pupils must:

1. Build a circuit with:
    - 6V supply
    - 15Ω resistor in series
    - Two 30Ω resistors in parallel
2. Complete a results table:

	Calculated	Simulated
Total Resistance (Ω)		
Circuit Current (A)		
Voltage across parallel section (V)		
Current in one 30Ω resistor (A)		

🧠 Higher Thinking Extensions

- Investigate what happens when a resistor in parallel is removed.
- Investigate what happens when resistance increases in series.
- Create a circuit that draws exactly 0.2A from a 12V supply.
- Build a circuit that divides voltage equally.
